

KINETICS & REACTOR DESIGN

CHE F311 · BITS Pilani · I-Semester 2026-27

Short Notes & Formula Sheet

Mole balances · Sizing · Rate laws · Isothermal design
Rate-data analysis · Multiple reactions · Non-isothermal design
Catalysis · Non-ideal reactors and residence-time distributions

Nine decks, 163 slides

Compiled from: the nine course slide decks in Slides/ — Ch1-krd (1).pdf (12 pp), Ch2-krd (1).pdf (15 pp), Ch3-krd.pdf (16 pp), Ch4-Krd.pdf (23 pp), Ch5-_krd.pdf (19 pp), Ch6_krd.pdf (18 pp), Ch7-_krd.pdf (24 pp), Catalysis_Krd.pdf (14 pp) and Non_ideal_KRD.pdf (22 pp).

Reference text: H. S. Fogler, *Elements of Chemical Reaction Engineering*; O. Levenspiel, *Chemical Reaction Engineering* (the decks' own problem sources).

Companion documents: KRD-Problems-and-Solutions.pdf works every numerical printed in these decks; krd-answers-check.py reproduces every number quoted below and exits 0.

One caveat about a source: Non_ideal_KRD.pdf is headed *Reaction Engineering / CHE G641* — the graduate version of this material. It is the only deck in the set that does not carry the CHE F311 heading, and section 9 says where that shows up.

What is inside

Topic	Deck
Mole balances, rate of reaction, conversion, the four design equations	Ch1
Reactor sizing from conversion; the Levenspiel plot; numerical integration; reactors in series and in parallel; optimum size ratio	Ch2
Rate laws and stoichiometry; relative rates; Arrhenius; stoichiometric tables for batch and flow, liquid and gas phase	Ch3
Isothermal reactor design; the algorithm; closed-form conversions; Damköhler number; pressure drop in packed beds	Ch4
Collection and analysis of rate data; differential and integral methods; half-life; differential reactor; nonlinear regression	Ch5
Multiple reactions; selectivity and yield; parallel and series schemes; product distribution	Ch6
Non-isothermal design; energy balance; adiabatic operation; heat exchange; multiple steady states and runaway; interstage cooling	Ch7
Catalysis and catalytic reactors; adsorption and site balance; Langmuir–Hinshelwood; internal and external diffusion	Catalysis
Non-ideal reactors; residence-time distribution; segregation; tanks-in-series; dispersion; two-parameter models	Non-ideal

Notation used throughout

X	conversion of A	τ	space time, V/v_0
$-r_A$	rate of disappearance of A	Da	Damköhler number
F_{A0}, F_A	molar flow of A in / out	k, A, E	rate constant, pre-exponential, activation energy
C_A, C_{A0}	concentration of A / at inlet	ε, δ	volume-change factor, mole-change number
V, W	reactor volume, catalyst mass	K_C, X_e	equilibrium constant, equilibrium conversion
N_A, N_{A0}	moles of A / initially	ΔH_{Rx}	heat of reaction
Θ_i	C_{i0}/C_{A0} , feed ratio	Ua, T_a	heat-transfer coefficient $\times$ area, coolant temperature
p	P/P_0 , reduced pressure	ϕ, η	Thiele modulus, effectiveness factor
α	pressure-drop parameter	$E(t), F(t)$	RTD function, cumulative RTD
t_m, σ^2	mean residence time, variance	Pe, n	Péclet number, tanks-in-series number

1 Mole Balances, Rates of Reaction and Conversion

Source: Ch1-krd (1).pdf.

1.1 Rate of reaction and the bases it can be written on

- **Rate of reaction** is the number of moles of a species reacting per unit time per unit *extensive quantity* of the system. The extensive quantity is what distinguishes the bases:

$-r_A = \text{mol}/(\text{volume} \cdot \text{time})$	homogeneous: per reactor volume V
$-r'_A = \text{mol}/(\text{mass cat.} \cdot \text{time})$	heterogeneous: per catalyst mass W
$r''_A = \text{mol}/(\text{area} \cdot \text{time})$	per catalyst surface area S
$-r'_A W = r''_A S = -r_A V$	$\Rightarrow$ all three give the same molar rate

Choosing the wrong basis is the most common error in packed-bed work: the design equation must be written on the same basis as the rate law.

1.2 Conversion, extent and the general mole balance

$X = \frac{\text{moles of A reacted}}{\text{moles of A fed}}$	conversion, defined on the limiting reactant
$N_i = N_i^0 + \nu_i \alpha$	extent of reaction α
$F_{j0} - F_j + G_j = \frac{dN_j}{dt}$	general mole balance on species j

- G_j is the rate of generation of j by reaction within the system volume — $G_j = r_j V$. Everything else in the balance is a transport term.
- Conversion is always defined on the **limiting** reactant, so that $0 \leq X \leq 1$.

1.3 The four design equations

Batch: $\frac{dN_A}{dt} = r_A V$	$\Rightarrow t = N_{A0} \int_0^X \frac{dX}{-r_A V}$
CSTR: $V = \frac{F_{A0} X}{(-r_A)_{\text{exit}}}$	rate evaluated at <i>exit</i> conditions
PFR: $\frac{dF_A}{dV} = r_A$	$\Rightarrow V = F_{A0} \int_0^X \frac{dX}{-r_A}$
PBR: $\frac{dF_A}{dW} = r'_A$	$\Rightarrow W = F_{A0} \int_0^X \frac{dX}{-r'_A}$

- The structural difference between a **CSTR** and a **PFR** is exactly the difference between evaluating $-r_A$ at the exit and integrating it along the path. That single fact drives all of section 2.
- Space time** $\tau = V/v_0$; **space velocity** $= v_0/V = 1/\tau$.

1.4 Worked from the deck

- Rocket engine** (slide 36). Chamber 75×60 cm, exhaust 108 kg/s.

$$V = 0.2121 \text{ m}^3 \quad \Rightarrow \quad \boxed{-r_{\text{H}_2} = 28 \text{ kmol m}^{-3} \text{ s}^{-1}, \quad -r_{\text{O}_2} = 14 \text{ kmol m}^{-3} \text{ s}^{-1}}$$

The 2:1 ratio of the two rates is the stoichiometry of $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ — the first appearance of the relative-rate rule developed in section 3.

- A→B at three duties** (slide 54). The deck asks for the volume of a CSTR and a PFR, and the batch time, at successively harsher conditions:

	(a)	(b)	(c)
CSTR	99.0 dm ³	2750 dm ³	66 000 dm ³
PFR	99.0 dm ³	127.9 dm ³	660 dm ³
Batch time	9.90 h	12.79 h	66.0 h

In (a) the two reactors are *equal* because the rate is constant there; from (b) onward the CSTR grows far faster than the PFR, which is the Levenspiel-plot argument of the next section in numerical form.

2 Conversion and Reactor Sizing

Source: Ch2-krd (1).pdf.

2.1 The Levenspiel plot

- Plot $F_{A0}/(-r_A)$ **against** X . Then the two design equations become areas on that single graph:

$$V_{\text{CSTR}} = \text{rectangle} = \left. \frac{F_{A0}}{-r_A} \right|_X \cdot X \quad \text{height at the exit} \times \text{width}$$

$$V_{\text{PFR}} = \text{area under the curve} = \int_0^X \frac{F_{A0}}{-r_A} dX \quad \text{integral from 0 to } X$$

- For any *positive-order* kinetics $-r_A$ falls as X rises, so $F_{A0}/(-r_A)$ rises and the **rectangle always exceeds the area under the curve**: a CSTR needs more volume than a PFR for the same conversion. The deck's own numbers in §1.4 show the gap opening from 0 to a factor of 100.

2.2 Numerical integration of the PFR integral

$$\begin{aligned} \text{Simpson (3 pt):} \quad \int &\approx \frac{h}{3} (f_0 + 4f_1 + f_2) \\ \text{Simpson (5 pt):} \quad \int &\approx \frac{h}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + f_4) \\ \text{3/8 rule:} \quad \int &\approx \frac{3h}{8} (f_0 + 3f_1 + 3f_2 + f_3) \end{aligned}$$

The five-point Simpson rule is the one the deck uses throughout, and it is the one that reproduces the printed answers.

2.3 Reactors in series

$$\begin{aligned} F_{Ai} &= F_{A0} (1 - X_i) && \text{flow leaving stage } i \\ V_i &= \frac{F_{A0} (X_i - X_{i-1})}{(-r_A)_i} && \text{CSTR } i : \text{rectangle from } X_{i-1} \text{ to } X_i \\ V_i &= F_{A0} \int_{X_{i-1}}^{X_i} \frac{dX}{-r_A} && \text{PFR } i : \text{area over the same interval} \end{aligned}$$

2.4 Reactors in parallel

- Rule:** for the total volume to be a minimum the feed must be split so that V/F (**equivalently** τ) **is the same in every branch**. Branches running at different τ waste volume.

2.5 Choosing the best arrangement

- Work from the $1/(-r_A)$ **versus** C_A curve.
- $n > 1$ (rate falls steeply): plug flow first, then a small mixed reactor, then a large mixed reactor.
- $n < 1$: the order is reversed — mixed flow first.
- Two CSTRs, minimum total volume:** maximise the inscribed rectangle, which gives the tangency condition $-dy/dx = y/x$ on the $1/(-r_A)$ curve.

2.6 Worked from the deck — the sizing table (slides 19–38)

The deck gives $T = 500$ K, $P = 830$ kPa, $F_{A0} = 0.4$ mol/s and this rate table. Note that the third row is $-r_A$ itself, not its reciprocal — the reciprocal is the last column.

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$-r_A$ (mol m ⁻³ s ⁻¹)	0.45	0.37	0.30	0.195	0.113	0.079	0.05
$F_{A0}/(-r_A)$ (m ³)	0.89	1.08	1.33	2.05	3.54	5.06	8.00

$$V_{\text{CSTR}} = 8.00 \times 0.8 = 6.40 \text{ m}^3$$

rectangle to $X = 0.8$

$$V_{\text{PFR}} = \frac{0.2}{3} (0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.00) = \boxed{2.165 \text{ m}^3}$$

5-pt Simpson

$$V_{\text{CSTR}} - V_{\text{PFR}} = 6.40 - 2.165 = 4.235 \text{ m}^3$$

the price of back-mixing

- **Two CSTRs in series** (slide 9), split at $X = 0.4$:

$$V_1 = 2.05 \times 0.4 = 0.82 \text{ m}^3, \quad V_2 = 8.00 \times (0.8 - 0.4) = 3.20 \text{ m}^3 \Rightarrow \boxed{4.02 \text{ m}^3}$$

- **Two PFRs in series** (slide 10), same split:

$$V_1 = 0.551 \text{ m}^3, \quad V_2 = 1.614 \text{ m}^3 \Rightarrow \boxed{2.165 \text{ m}^3} \text{ (identical to one PFR)}$$

Two points worth keeping. First, splitting a PFR changes nothing — the integral is additive — whereas splitting a CSTR cuts the volume from 6.40 to 4.02 m³, a 37 % saving. Second, the series-CSTR total still exceeds the PFR, and keeps falling as more stages are added, approaching the PFR area as $N \rightarrow \infty$.

- **Unknown stoichiometry from an MFR** (slide 42). $C_{A0} = 0.10$, $C_{B0} = 0.01$ mol/L, $V = 1$ L, outlet 0.02/0.03/0.04:

$$r_A = -0.08, \quad r_B = +0.02, \quad r_C = +0.04 \Rightarrow \boxed{4A \rightarrow B + 2C}$$

- **Three PFRs in two parallel branches** (slide 46). Branch D holds 50 + 30 L, branch E holds 40 L; equal τ requires

$$\text{fraction to D} = \frac{50}{90} = \boxed{0.556}$$

- **P2-7** (slide 59), $1/(-r_A) = 1.0, 1.67, 5.0, 5.0, 5.0, 5.0, 1.25, 0.91$ at $X = 0, 0.2, 0.4, 0.45, 0.5, 0.6, 0.8, 0.9$:

-
- (a) CSTR 24.0 dm³; PFR 71.9 dm³ for $X = 0.8$
 - (b) the two volumes are equal at $X \approx 0.69$
 - (c) a 105 dm³ CSTR gives $X = 0.70$
 - (d) CSTR then PFR: $X_1 = 0.401$, $X_2 = 0.643$
 - (e) PFR then CSTR: $X_1 = 0.400$, $X_2 = 0.900$
-

(d) and (e) are the same two reactors in the opposite order and differ by 0.26 in exit conversion — the $1/(-r_A)$ curve here is *non-monotonic*, so the ordering rule of §2.5 actually bites.

3 Rate Laws and Stoichiometry

Source: Ch3-krd.pdf.

3.1 Relative rates of reaction



$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d} \Rightarrow \boxed{r_B = \frac{b}{a} r_A}$$

Divide each species' rate by its own stoichiometric coefficient *with sign*; the results are all equal. This is the check that closes the rocket-engine problem of §1.4.

3.2 Form of the rate law

$$-r_A = k C_A^\alpha C_B^\beta$$

power law; overall order $n = \alpha + \beta$

$$k = A e^{-E/RT}$$

Arrhenius

$$\ln \frac{k_2}{k_1} = \frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

two-temperature form

$$K_C = \frac{k_f}{k_r}$$

elementary reversible; $-r_A \rightarrow 0$ at equilibrium

- α and β are empirical — they are *not* the stoichiometric coefficients unless the reaction is elementary. Determining them is the whole subject of section 5.

3.3 Stoichiometric tables

For $aA + bB \rightarrow cC + dD$, with $\Theta_i = C_{i0}/C_{A0}$.

Batch	$N_A = N_{A0}(1 - X), \quad N_B = N_{A0}(\Theta_B - \frac{b}{a}X), \quad N_C = N_{A0}(\Theta_C + \frac{c}{a}X)$
Batch, liquid	$V = V_0$, so $C_i = N_i/V_0$
Flow	the same expressions with $N \rightarrow F$
Batch, gas, const. P, T	$V = V_0(1 + \varepsilon X), \quad C_A = \frac{N_{A0}(1 - X)}{V_0(1 + \varepsilon X)}$

$$C_i = C_{A0} \frac{\Theta_i + \nu_i X/a}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0} \quad \text{gas-phase flow, general}$$

$$\varepsilon = y_{A0} \delta, \quad \delta = \frac{c + d - b - a}{a} \quad \text{mole change on complete conversion}$$

- $\varepsilon = 0$ whenever there is no change in mole number, or the feed is diluted enough that y_{A0} is small. Setting $\varepsilon = 0$ is the single most common legitimate simplification, and the deck's gas-phase examples all state it explicitly.
- $\varepsilon > 0$ (moles increase) means the gas expands as conversion proceeds, concentrations fall faster than stoichiometry alone implies, and the reactor needs more volume than the constant-density calculation suggests.

3.4 Worked from the deck

- **Relative rates** (slides 5–7).

$$\begin{aligned} r_{\text{NO}_2} = +4 &\implies r_{\text{NO}} = -4, \quad r_{\text{O}_2} = -2 \text{ mol m}^{-3}\text{s}^{-1} \\ r_A = -10 &\implies r_B = -15, \quad r_C = +25 \text{ mol dm}^{-3}\text{s}^{-1} \end{aligned}$$

- **Dimerisation in an MFR** (slide 61). Four runs give

$$X = 0.2502, 0.4996, 0.6667, 0.7995; \quad -r_A = 2502.2, 1498.9, 800.0, 399.8 \text{ mol L}^{-1}\text{h}^{-1}$$

$$\boxed{n = 1.96 \approx 2}, \quad k = 0.3407, 0.3369, 0.3200, 0.3583 \text{ L mmol}^{-1}\text{h}^{-1}$$

The scatter in k across the four runs — 0.320 to 0.358, about $\pm 6\%$ — is the practical test that the assumed order is right: a wrong order makes k drift systematically instead of scattering.

- **Phosphine decomposition** (slide 63). $4\text{PH}_3 \rightarrow \text{P}_4 + 6\text{H}_2$, first order, $k = 10/\text{h}$, 649°C , 460 kPa , 40 mol/h , 80% conversion:

$$C_{A0} = 60.00 \text{ mol m}^{-3} \implies \boxed{V = 0.1882 \text{ m}^3 \approx 188 \text{ L}}$$

C_{A0} comes from P/RT at the stated temperature and pressure — the deck supplies the state, not the concentration.

4 Isothermal Reactor Design

Source: Ch4-Krd.pdf.

4.1 The algorithm

Five steps, in this order, every time.

1. **Mole balance** — pick the reactor, write its design equation (section 1).
2. **Rate law** — from the data or the mechanism (section 3).
3. **Stoichiometry** — express every C_i in terms of X (the tables of §3.3).
4. **Combine** — substitute 2 and 3 into 1 to get one equation in X .
5. **Evaluate** — integrate, or solve, and size the reactor.

Every worked example in this section follows those five steps; the deck labels them explicitly on the slides.

4.2 Closed-form conversions

Batch, 1st order	$t = \frac{1}{k} \ln \frac{1}{1-X}$
Batch, 2nd order	$t = \frac{1}{kC_{A0}} \cdot \frac{X}{1-X}$
CSTR, 1st order	$X = \frac{k\tau}{1+k\tau}$
CSTR, 2nd order	$X = \frac{(1+2Da) - \sqrt{1+4Da}}{2Da}, \quad Da = k\tau C_{A0}$

$Da_1 = k\tau$	Damköhler number, first order
$Da \leq 0.1 \implies X < 0.1$	reaction-limited: little conversion
$Da \geq 10 \implies X > 0.9$	near-complete conversion

Da is the ratio of the reaction rate to the convective transport rate. It is the quickest way to see, before any integration, whether a proposed reactor is remotely big enough.

4.3 Reactors in series and in parallel

$$N \text{ equal CSTRs, 1st order: } X = 1 - \frac{1}{(1+k\tau_i)^N}$$

Parallel CSTRs: behave as one large CSTR of the same total volume

Compare with §2.6: there, splitting the duty into two CSTRs saved 37% of the volume. This is the same result in closed form, and it is why industrial trains are staged rather than single-vessel.

4.4 Variable volume, gas phase

$$V_{\text{PFR}} = \frac{v_0}{kC_{A0}} \left[2\varepsilon(1+\varepsilon) \ln(1-X) + \varepsilon^2 X + \frac{(1+\varepsilon)^2 X}{1-X} \right] \quad \text{2nd order, gas phase}$$

4.5 Pressure drop in packed beds

$$\begin{aligned} \frac{dp}{dW} &= -\frac{\alpha}{2p}, & p &= \frac{P}{P_0} & \text{reduced-pressure form} \\ \varepsilon = 0: \quad p &= (1 - \alpha W)^{1/2} & & \text{closed form} \\ C_A &= C_{A0} \frac{1-X}{1+\varepsilon X} p & & \text{gas, constant } T \\ \frac{dP}{dz} &= -\frac{\beta_0 P_0}{P} (1 + \varepsilon X), & \beta_0 &= \frac{G(1-\phi)}{\rho_0 D_p \phi^3} \left[\frac{150(1-\phi)\mu}{D_p} + 1.75 G \right] \end{aligned}$$

- ϕ is the bed porosity, G the superficial mass flux, D_p the particle diameter. β_0 is the **Ergun equation** in the form the deck uses.
- Because C_A carries a factor p , pressure drop *slows the reaction*, so the conversion achieved in a given catalyst mass is lower than the no-drop calculation predicts. The two must be solved **coupled**, not sequentially.

4.6 Worked from the deck

- **Ethylene oxide, batch** (slides 17–20). Regression of the concentration–time data gives

$$k = 0.315 \text{ min}^{-1}, \quad t_{90\%} = 7.32 \text{ min}$$

and, at the much slower $k = 10^{-4} \text{ s}^{-1}$,

$$t_{90\%} = \boxed{6.40 \text{ h}}$$

- **Ethylene glycol, CSTR** (slides 29–38):

$$F_{A0} = 7.672 \text{ lbmol/min}, \quad v_0 = 15.34 \text{ ft}^3/\text{min}, \quad \boxed{V_{80\%} = 197.3 \text{ ft}^3}$$

Arrangement	τ (min)	Da	X
two CSTRs in parallel	13.94	4.34	0.813
two CSTRs in series	6.97	2.168	0.900

Same total volume, and the series arrangement reaches 0.900 against 0.813 — the staging argument of §4.3 with the deck's own numbers.

- **Ethylene cracking, PFR** (slide 44). 300 M lb/yr, 1100 K, 6 atm, $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82 \text{ kcal/mol}$:

$$k(1100 \text{ K}) = 3.067 \text{ s}^{-1}, \quad F_{A0} = 154.1 \text{ mol/s}, \quad C_{A0} = 66.476 \text{ mol/m}^3$$

$$v_0 = 2.318 \text{ m}^3/\text{s} \quad \Rightarrow \quad \boxed{V = 1.83 \text{ m}^3 = 64.6 \text{ ft}^3}$$

3150 tubes of 0.0205 ft^3 each. The Arrhenius step is what moves k from 0.072 to 3.067 s^{-1} — a factor of 43 for 100 K.

- **Packed bed**, $A + B \rightarrow 2C$ (slides 55–58), $\alpha = 0.0099$, $W = 100$:

$$\boxed{X = 0.602 \text{ with pressure drop}} \quad \text{vs.} \quad \boxed{X = 0.75 \text{ without}}$$

Ignoring pressure drop overstates conversion by 0.15, a quarter of the achieved value. This is the number to quote when justifying a coupled solution.

- **PBR and CSTR in either order** (slide 76):

Arrangement	group	X
PBR then CSTR	0.02471	0.598
CSTR then PBR	0.02471	0.690
PBR then CSTR, $\alpha' = 0.00711$	0.02471	0.727

Exit $p = 0.3162$, i.e. $P = 6.32 \text{ atm}$. Putting the CSTR *first* gains 0.09 in conversion, because the CSTR then operates at full pressure and the PBR only sees the already-reduced rate. Reducing α' recovers most of the loss — pressure drop, not ordering, is the dominant effect.

- **Series reactions in batch**, $A \rightarrow B \rightarrow C$ (slide 78):

$$t_{\max} = 0.693 \text{ h}, \quad \boxed{C_{B,\max} = 1.250}$$

- **Pressure-drop length check** (slides 90–98). Without pressure drop, 16.8 m reaches $X = 0.10$ and 35.5 m reaches $X = 0.20$.

A discrepancy in the source, reported rather than smoothed over. The deck states that about 20 m achieves the target *with* pressure drop, and asks for $X = 0.20$ in a 1.5 cm bed. The no-drop calculation already needs 35.5 m for $X = 0.20$, and adding pressure drop can only make the bed longer — so $X = 0.20$ is not reachable in 1.5 cm. The deck's own figures are reproduced above and the conflict is flagged here rather than resolved by adjusting a number.

5 Collection and Analysis of Rate Data

Source: Ch5-_krd.pdf.

5.1 Differential method

- Take logs of the power law to linearise it:

$$\ln\left(-\frac{dC_A}{dt}\right) = \ln k + \alpha \ln C_A \quad \text{slope} = \alpha, \text{ intercept} = \ln k$$

The derivatives come from the concentration–time data by finite differences:

initial point	$\frac{-3C_0 + 4C_1 - C_2}{2\Delta t}$
interior point	$\frac{C_{i-1} - C_{i+1}}{2\Delta t}$
final point	$\frac{C_{N-2} - 4C_{N-1} + 3C_N}{2\Delta t}$

5.2 Integral method

- **Guess the order, integrate, and test for a straight line.** The order that linearises the data is the order.

order 0	C_A against t is linear
order 1	$\ln C_A$ against t is linear
order 2	$1/C_A$ against t is linear
$A + B, M = C_{B0}/C_{A0} \neq 1$	$\ln \frac{C_A}{C_B} = \ln M + C_{A0}(M - 1)kt$

5.3 Half-life method

$$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)k} C_{A0}^{1-n} \quad \text{order } n \neq 1$$

$$\ln t_{1/2} = (1-n) \ln C_{A0} + \text{const} \quad \text{slope} = 1-n \implies n = 1 - \text{slope}$$

$$n = 1: \quad t_{1/2} = \frac{\ln 2}{k} \quad \text{independent of } C_{A0}$$

A half-life that does not change with initial concentration is the signature of first-order kinetics — no regression needed.

5.4 Differential reactor and initial rates

$$-r'_A = \frac{F_{A0} - F_A}{W} \approx \frac{F_P}{W} \quad \text{low conversion, so composition is nearly uniform}$$

5.5 Nonlinear regression and enzyme kinetics

$$s^2 = \sum_i (C_{m,i} - C_{c,i})^2 \quad \text{minimise; } m = \text{measured, } c = \text{calculated}$$

$$-r_S = \frac{k_3 C_S C_E}{C_S + C_M} \quad \text{Michaelis-Menten}$$

$$k_3 C_E t = (C_{S0} - C_S) + C_M \ln \frac{C_{S0}}{C_S} \quad \text{integrated form}$$

5.6 Variable-volume batch, constant pressure

$$X = \frac{(V/V_0) - 1}{\varepsilon_A} \quad \text{read conversion straight off the volume change}$$

5.7 Worked from the deck

Problem	Result
Trityl + methanol (regression of $\ln t_{1/2}$)	$n = 2.14$; $k' = 0.1243$; $k = 0.249 \text{ dm}^6 \text{ mol}^{-2} \text{ min}^{-1}$
Dimethyl ether, constant volume	$p_{i0} = 310.3$; $k = 4.4, 4.3, 4.4, 4.9 \times 10^{-4} \text{ s}^{-1}$, mean 4.50×10^{-4}
$A \rightleftharpoons R$, reversible	$k_1 = 0.05776$, $k_2 = 0.02888$
L5.19	$n = 2.00$; $k = 0.0040, 0.0040, 0.00402, 0.00401 \text{ L mmol}^{-1} \text{ min}^{-1}$
L3.25	$k = 0.1604 \text{ min}^{-1}$
Bomb reactor, $2A \rightarrow B$	1st order $k = 0.0517$, $r^2 = 0.886$; 2nd order $k = 0.0741$, $r^2 = 0.964$
Six MFRs in series vs one PFR	$X_{\text{MFRs}} = 0.750$, $X_{\text{PFR}} = 0.667$
Two-reactor split	$X_A = 0.2308$, $X_B = 0.1154$, $C_B = 46.00$
Sucrose inversion, Michaelis-Menten	$k_3 C_{S0} = 0.3210 \text{ h}^{-1}$; $C_M = 0.5151 \text{ mmol/L}$; $k_3 = 32.1 \text{ h}^{-1}$

- **The bomb-reactor comparison is the cleanest illustration of the integral method in the whole deck.** Both orders were fitted to the same data; the second-order fit has $r^2 = 0.964$ against 0.886 for first order. The higher r^2 is what selects the order — not a preference, and not the value of k .
- **Reversible PFR** (slide 61), $A \rightleftharpoons R$:

$$X_e = 0.80, \quad \tau = 20.0 \text{ min}, \quad C_{Af} = 0.0494 \quad \implies \quad \boxed{X = 0.506}$$

$X = 0.506$ against an equilibrium ceiling of $X_e = 0.80$: the reactor is approaching equilibrium, not running out of residence time. Adding volume beyond this point buys very little, which is the practical significance of computing X_e first.

6 Multiple Reactions

Source: Ch6_krd.pdf.

6.1 Selectivity and yield

$S_{D/U} = \frac{r_D}{r_U}$	instantaneous selectivity
$\tilde{S}_{D/U} = \frac{F_D}{F_U}$	overall (batch: N_D/N_U)
$\tilde{Y}_D = \frac{r_D}{-r_A}$	instantaneous yield
$\tilde{Y}_D = \frac{F_D}{F_{A0} - F_A}$	overall yield
$r_j = \sum_i \nu_{ji} r_i$	net rate of j over all reactions

6.2 Parallel reactions — the operating rules

For $A \xrightarrow{k_1} D$ and $A \xrightarrow{k_2} U$ with $-r_{1A} = k_1 C_A^{\alpha_1}$ and $-r_{2A} = k_2 C_A^{\alpha_2}$, let $a = \alpha_1 - \alpha_2$.

$a > 0$	keep C_A high — PFR or batch, high pressure, no inerts
$a < 0$	keep C_A low — CSTR, add inerts, dilute the feed
$E_D > E_U$	run hot
$E_U > E_D$	run cool , but still fast enough to be practical

- **Two reactants:** the signs of $(\alpha_1 - \alpha_2)$ and $(\beta_1 - \beta_2)$ together choose the contacting pattern — semibatch, side streams, or a train of small CSTRs.

6.3 Series reactions

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \quad \implies \quad t_{\text{opt}} = \frac{\ln(k_2/k_1)}{k_2 - k_1} \quad \text{maximises } C_B$$

Verified numerically in §4.6: $t_{\text{max}} = 0.693$ h gives $C_{B,\text{max}} = 1.250$.

6.4 Product distribution

$$\varphi\left(\frac{R}{A}\right) = \frac{dC_R}{-dC_A} \quad \text{instantaneous fractional yield}$$

$$C_R = \int \varphi(-C_A) \quad \text{PFR: area under } \varphi; \quad \text{MFR: rectangle; staged: steps}$$

6.5 Worked from the deck

- **Product distribution** (slide 32). The same feed through three contacting patterns:

$$(a) C_R = 2.859 \quad (b) C_R = 4.50 \quad (c) C_R = 7.295$$

A factor of 2.6 in product concentration from contacting alone, with no change to chemistry, temperature or throughput. This is why the rules of §6.2 are worth memorising.

- **Two MFRs in series** (slide 33), $k_1 = 0.5$, $k_2 = 0.4$, $\tau = 2.5$ and 10 min:

$$C_{A2} = 0.0745, \quad C_{R2} = 0.2277, \quad C_{S2} = 0.9978$$

7 Non-Isothermal Reactor Design

Source: Ch7-_krd.pdf.

7.1 The energy balance

$$\dot{Q} - \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = 0 \quad \text{steady state, no shaft work beyond } \dot{W}_s$$

$$H_i(T) = H_i^\circ(T_R) + C_{p,i}(T - T_R) \quad \text{enthalpy referenced to } T_R$$

7.2 Adiabatic operation and the equilibrium ceiling

$$X = \frac{\sum \Theta_i C_{p,i} (T - T_0)}{-[\Delta H_{Rx}^\circ(T_R) + \Delta C_p (T - T_R)]} \quad \text{adiabatic energy balance}$$

$$T = T_0 + (\text{slope}) X \quad \text{a straight line on the } T\text{-}X \text{ plane}$$

$$K_C(T) = K_C(T_1) \exp\left[\frac{\Delta H_{Rx}}{R} \left(\frac{1}{T_1} - \frac{1}{T}\right)\right] \quad \text{van 't Hoff}$$

$$X_e = \frac{K_C}{1 + K_C} \quad \text{for } A \rightleftharpoons B$$

- For an **exothermic reversible** reaction these two lines move in opposite directions: temperature raises the rate but *lowers* X_e . The achievable conversion is where the energy-balance line meets the equilibrium curve, and no amount of catalyst moves that point.

7.3 Heat exchange in a tubular reactor

$$\frac{dT}{dV} = \frac{Ua(T_a - T) + (-r_A)(-\Delta H_{Rx})}{\sum F_i C_{p,i}}$$

co-current coolant: $\frac{dT_a}{dV} = \frac{Ua(T - T_a)}{\dot{m}_c C_{p,c}}$

7.4 CSTR with cooling, and multiple steady states

$$G(T) = \frac{-\Delta H_{Rx}}{1 + 1/(k\tau)} \quad \text{heat generated}$$

$$R(T) = C_{p0}(1 + \kappa)(T - T_c) \quad \text{heat removed; } \kappa \propto UA/F_{A0}$$

$$S' = \frac{E}{RT_s^2} G(T_s) - C_{p0}(1 + \kappa) \quad \text{stability at the steady state } T_s$$

- Intersections of $G(T)$ and $R(T)$ are the steady states. Where $S' < 0$ the state is stable; where $S' > 0$ it is not.
- Runaway** occurs when the two curves become tangent and the intersection disappears — raising T_c or lowering the feed temperature moves the system back to a single stable state. This is the **ignition/extinction** behaviour the deck plots on its stability diagram.

7.5 Interstage cooling

- Instead of one adiabatic bed taken to its equilibrium limit, use **several adiabatic stages with cooling between them**. Each stage starts cooler, so each one starts further from equilibrium and gains more conversion. The optimum feed temperature for a fixed total volume maximises exit X .

7.6 Worked from the deck

- Butane isomerisation, adiabatic PFR** (slides 14–16). The energy balance gives $T = 330 + 43.3X$, i.e. 6900/158.9, with $E/R \approx 7906$ and $-\Delta H/R \approx 830.3$; k rises from 4.22 at 330 K to 31.67 at 360.3 K.

$$V_{X=0.70} = 1.974 \text{ m}^3 \quad \text{deck table: } T(X=0.2) = 338.7 \text{ K}, \quad (-r_A)_{X=0} = 39.3$$

- Interstage cooling** (slides 48–51), 40 mol/s cooled to 350 K between stages, target 95% of X_e :

$$\text{R1: } X = 0.397 \text{ at } 458.9 \text{ K}; \quad \text{R2: } X = 0.621 \text{ at } 439.5 \text{ K}$$

$$\dot{Q}_1 = 217.8 \text{ kcal/s}, \quad \dot{Q}_2 = 179.1 \text{ kcal/s}$$

The second stage exits *cooler* than the first (439.5 vs 458.9 K) yet reaches *higher* conversion — the direct payoff of resetting the distance to equilibrium.

- **Propylene glycol, CSTR** (slides 57–74):

Case	X	T (R)	T (F)
adiabatic	0.842	610.8	151
with cooling coil	0.3454	562.18	102.2

The deck’s conclusion, in its own words: “*125 F = 585 F — we can’t use the existing system.*” The adiabatic operating point sits above the temperature the plant can accept, and the coil pulls conversion down to 0.345 to hold it. Note that this is the same system as §5.7’s Michaelis–Menten-free propylene-glycol example; the cross-over is at 611 R = 151 °F, $X \approx 0.84$.

- **Parallel reactions with heat effects** (slide 95). The ODE set integrated to $V = 10^4 \text{ dm}^3$ gives $T = 645 \text{ K}$.

8 Catalysis and Catalytic Reactors

Source: `Catalysis_Krd.pdf`.

8.1 The seven steps and the rate-limiting-step algorithm

External diffusion → internal diffusion → **adsorption of A** → surface reaction → **desorption of B** → internal diffusion out → external diffusion out. Assume one step limits, write its rate, and combine with the **site balance**

$$C_t = C_v + \sum_i C_{i \cdot S}$$

Then test the assumption: if the assumed rate-limiting step does not reproduce the data, move to the next one. The deck works through each assumption in turn.

8.2 Adsorption isotherms and rate forms

$$C_{A \cdot S} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B}$$

Langmuir, competitive sites

$$-r'_T = \frac{k P_T P_{H_2}}{1 + K_B P_B + K_T P_T}$$

Eley–Rideal type

- **Langmuir–Hinshelwood** assumes both reactants are adsorbed before reacting; **Eley–Rideal** assumes one reacts directly from the gas phase. The denominator is the site balance in both cases — that is what makes the rate saturate at high pressure.

8.3 Internal diffusion — Thiele modulus and effectiveness factor

$$\phi_s = R \sqrt{\frac{k_1}{D_e}}$$

Thiele modulus, first order, sphere

$$\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1)$$

effectiveness factor, sphere

$$\eta = \frac{\tanh \phi}{\phi}$$

slab geometry

$$\frac{C_A}{C_{As}} = \frac{R}{r} \cdot \frac{\sinh(\phi r/R)}{\sinh \phi}$$

concentration profile inside the pellet

- $\eta \rightarrow 1$ for small ϕ (reaction-limited: the whole pellet is used); $\eta \rightarrow 3/\phi$ for large ϕ (diffusion-limited: only a shell near the surface reacts).

8.4 Diagnostic criteria

$C_{WP} = \eta \phi^2 = \frac{(-r'_A)_{\text{obs}} R^2}{D_e C_{As}}$	Weisz-Prater
$C_{WP} \ll 1$	no internal limitation
$C_{WP} \gg 1$	strongly limited
$\Omega = \frac{\eta}{1 + \eta \phi^2 S_a \rho_b / k_c}$	overall effectiveness, with external resistance
$\frac{k C_{Ab}^{n-1} R^2 \rho_b S_a E}{b D_e T^2} < 0.15$	Mears: external gradients negligible

Weisz-Prater is the practical one: it uses the *observed* rate, so it can be evaluated from experiment without knowing the intrinsic kinetics.

8.5 Worked from the deck

- **Concentration inside a pellet** (slide 48), sphere $d_p = 2 \times 10^{-3}$ cm, $D_e = 0.1$ cm²/s:

$$\phi = 5.986, \quad \frac{C_A}{C_{As}} = 0.2371, \quad C_A = 2.37 \times 10^{-4}$$

$$\text{to reach } C_A/C_{As} = 0.8 : \quad \phi = 2.042 \quad \implies \quad \boxed{d_{\text{new}} = 6.82 \times 10^{-4} \text{ cm}}$$

At $\phi = 5.99$ only 24 % of the surface concentration survives at the pellet centre. Shrinking the diameter to about a third of the original is what it takes to make the whole pellet effective.

- **Apparent activation energy** (slide 57), $R = 0.2$ cm, $D_e = 0.015$ cm²/s, $100 \rightarrow 150$ °C:

T	ϕ	η	r
100 °C	1.575	0.8660	2.617×10^{-5}
150 °C	7.75	0.337	2.468×10^{-4}

$$\boxed{E_{\text{app}} = 14\,080 \text{ cal/mol}}$$

The measured activation energy is roughly *half* the true one because η is falling as fast as k is rising — the classic signature of internal diffusion control. Halving the observed E is the diagnostic, not an error in the Arrhenius plot.

- **Reactor choice:** the packed bed is the workhorse; the deck covers adiabatic and non-adiabatic operation, reverse-flow and counter-flow configurations.

9 Non-Ideal Reactors and Residence-Time Distributions

Source: Non_ideal_KRD.pdf — headed *Reaction Engineering / CHE G641*, the graduate treatment of this material.

9.1 The RTD functions

$E(t) = \frac{C(t)}{\int_0^\infty C dt}$	normalised tracer response
$F(t) = \int_0^t E dt$	cumulative fraction younger than t
$t_m = \int_0^\infty t E dt = \frac{V}{v_0}$	mean residence time
$\sigma^2 = \int_0^\infty (t - t_m)^2 E dt$	variance, i.e. the spread

Normalising by the area is what makes $E(t)$ a probability density, and it is the step most often missed when a pulse experiment is worked up by hand.

9.2 Ideal limits

PFR: $E(t) = \delta(t - \tau)$	all fluid elements the same age
CSTR: $E(t) = \frac{e^{-t/\tau}}{\tau}$	exponential; some fluid leaves almost at once

9.3 Segregation model

$$\bar{X} = \int_0^\infty X_{\text{batch}}(t) E(t) dt \quad \text{each fluid element is a small batch reactor}$$

- $n > 1$: $\bar{X}$ exceeds the ideal-mixed conversion. $0 < n < 1$: the opposite. Complete segregation is therefore a bound, and it tells you which way a real reactor departs from ideal mixing without any further modelling.

9.4 Tanks-in-series model

$$E(t) = \frac{t^{n-1}}{(n-1)! \tau_i^n} e^{-t/\tau_i} \quad \text{gamma distribution}$$

$$n = \frac{t_m^2}{\sigma^2} \quad \text{number of tanks, from the tracer data alone}$$

$$X = 1 - (1 + k\tau_i)^{-n} \quad \text{first-order conversion}$$

9.5 Dispersion model

$$X = 1 - \frac{4q e^{Pe/2}}{(1+q)^2 e^{Pe q/2} - (1-q)^2 e^{-Pe q/2}} \quad \text{closed vessel, first order}$$

$$q = \sqrt{1 + \frac{4Da}{Pe}}, \quad Da = k\tau, \quad Pe = \frac{UL}{Da}$$

$$\sigma_\theta^2 = \frac{2}{Pe} - \frac{2}{Pe^2} (1 - e^{-Pe}) \quad \text{link from tracer data to } Pe$$

$$n = \frac{Pe}{2} + 1 \quad \text{equivalent number of tanks}$$

9.6 Two-parameter models

- A real vessel is represented as an **ideal region** of volume $V_s = \alpha V$ exchanging with a **dead zone**, plus a **bypass** fraction $\beta = v_b/v_0$.
- Tracer response takes the form

$$\frac{C_T}{C_{T0}} = 1 - (1 - \beta) e^{-(1-\beta)t/(\alpha\tau)}$$

and the two parameters (α, β) are extracted from the first two moments.

9.7 Worked from the deck

- **Pulse tracer** (slide 17):

$$\text{area} = 50.03, \quad t_m = 5.19 \text{ min}, \quad \sigma^2 = 6.03 \quad \Rightarrow \quad \boxed{\text{fraction between 3 and 6 min} = 0.513}$$

Just over half the fluid leaves within a minute either side of the mean — the spread is the whole story here, not the mean.

- **Dispersion example** (slides 52–59):

$$Pe = 7.74, \quad Da = 1.28$$

Model	$\bar{X}$
dispersion (closed vessel)	0.679
tanks-in-series, $n = 4.45$	0.676
ideal PFR	0.720
ideal CSTR	0.560

The two real-reactor models agree to 0.003 and sit *between* the ideal limits, as they must. Using $n = Pe/2 + 1 = 4.87$ instead of the moment-derived 4.45 changes the answer negligibly — but the agreement of the two independent models is what justifies trusting either. Treating this vessel as an ideal CSTR would understate conversion by 0.12; as an ideal PFR, overstate it by 0.04.

Formula Sheet — the whole course on one page

$X = \frac{N_{A0} - N_A}{N_{A0}}$	$\tau = \frac{V}{v_0}$
$V_{\text{CSTR}} = \frac{F_{A0}X}{(-r_A)_{\text{exit}}}$	$V_{\text{PFR}} = F_{A0} \int_0^X \frac{dX}{-r_A}$
$W_{\text{PBR}} = F_{A0} \int_0^X \frac{dX}{-r'_A}$	$t_{\text{batch}} = N_{A0} \int_0^X \frac{dX}{-r_A V}$
$-r_A = k C_A^\alpha C_B^\beta$	$k = A e^{-E/RT}$
$C_i = C_{A0} \frac{\Theta_i + \nu_i X/a}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0}$	$\varepsilon = y_{A0} \delta$
$X_{\text{CSTR,1st}} = \frac{k\tau}{1 + k\tau}$	$X_{N \text{ CSTRs}} = 1 - \frac{1}{(1 + k\tau_i)^N}$
$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)k} C_{A0}^{1-n}$	$\ln t_{1/2} = (1-n) \ln C_{A0} + \text{const}$
$S_{D/U} = \frac{r_D}{r_U}$	$\tilde{Y}_D = \frac{F_D}{F_{A0} - F_A}$
$T = T_0 + (\text{slope})X$	$X_e = \frac{K_C}{1 + K_C}$
$\phi = R\sqrt{k_1/D_e}$	$\eta = \frac{3}{\phi^2}(\phi \coth \phi - 1)$
$C_{WP} = \frac{(-r'_A)_{\text{obs}} R^2}{D_e C_{As}}$	$n = \frac{t_m^2}{\sigma^2} = \frac{Pe}{2} + 1$

The four comparisons worth knowing cold

Question	Better	From
Same conversion, one vessel: PFR or CSTR?	PFR	§2.6 (2.165 vs 6.40 m ³)
Same total volume, two CSTRs in series or parallel?	series	§4.6 (0.900 vs 0.813)
PBR then CSTR, or CSTR then PBR?	CSTR first	§4.6 (0.690 vs 0.598)
Real vessel, or an ideal limit?	neither	§9.7 (0.679 vs 0.720 / 0.560)

Prepared from the nine CHE F311 slide decks named in the box on page 1; every formula above is the form taught in the deck heading its section, and every number is reproduced by `krd-answers-check.py`, which exits 0. Structure and depth follow the course standard set by `SP2-notes.tex` (CHE F313). Two places where the source itself is inconsistent are reported rather than corrected: the pressure-drop length in §4.6, and the graduate (CHE G641) heading on `Non_ideal_KRD.pdf`. Symbols: X conversion, $-r_A$ rate of disappearance of A, $-r'_A$ rate per catalyst mass, F_{A0} inlet molar flow of A, C_A concentration, Θ_i feed ratio C_{i0}/C_{A0} , ε volume-change factor, δ mole-change number, V volume, W catalyst mass, τ space time, Da Damköhler number, k rate constant, A pre-exponential factor, E activation energy, K_C equilibrium constant, X_e equilibrium conversion, ΔH_{Rx} heat of reaction, Ua heat-transfer coefficient times area, T_a coolant temperature, p reduced pressure P/P_0 , α pressure-drop parameter, ϕ bed porosity or Thiele modulus as the section states, η effectiveness factor, D_e effective diffusivity, C_{WP} Weisz–Prater number, $E(t)$ residence-time distribution, $F(t)$ cumulative RTD, t_m mean residence time, σ^2 variance, Pe Péclet number, n tanks-in-series number, $S_{D/U}$ selectivity, $\tilde{Y}_D$ yield.